

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

MAY EXAMINATION

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS

TUESDAY, 3 JUNE 2025

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATION AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Candidates are not allowed to use any calculator.

Answer questions only in the answer booklet provided.

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION A**

- Q1. (a) Propose an Amazon Web Services (AWS) architecture diagram in scaling your cloud infrastructure. Your design should consider:
- Choose an Amazon Machine Image (AMI) for your running web instance(s).
 - Setup relational database(s) in storing web data.
 - Perform application load balancing.
 - Create an auto scaling group between availability zones.
 - Automatically scale new instances within a private subnet.
 - Allow your cloud resources in the private subnet in accessing internet.
 - Create alarms and monitor performance of your infrastructure, and
 - Integrating some relevant intelligent cloud services.

(10 marks)

- (b) You hosted an application on AWS that lets users to render images and do some general computing. Which of the below listed services can be used to route the incoming user traffic? Justify your answer. (4 marks)

- Classic Load Balancer
- Application Load Balancer
- Network Load balancer

- (c) AWS offers several compute and database options to meet different needs. Which AWS service(s) can be used to host an off-premises MySQL database with minimal administrative operations? Justify your answer. (4 marks)

- (d) Answer the following questions with short justification in the context of AWS.

- (i) Which AWS service automatically executes code in response to events from services like S3, DynamoDB, and API Gateway? (2 marks)
- (ii) Which AWS service enables you to create an isolated network environment for your cloud resources? (2 marks)
- (iii) An external audit is being conducted on UTAR. The IT auditor requires a log of all accesses to the AWS resources in UTAR's account. Which AWS cloud service can assist in providing these details? (3 marks)

[Total : 25 marks]

- Q2. (a) From your point of view, how does ChatGPT utilize cloud computing technology to enhance its natural language processing capabilities? Please provide **THREE (3)** points to support your justification. (6 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q2. (Continued)**

- (b) What are the key differences between grid computing and cloud computing in terms of **resource allocation, ownership, and usage scenarios**? (6 marks)
- (c) Compare and contrast the cluster computing architecture with the peer-to-peer architecture in the context of distributed systems. Discuss the key differences in their **design, functionalities, security, and scalability**. Present your answer in a table. (8 marks)
- (d) Define microservices and present **TWO (2)** advantages of microservices over traditional web services. (5 marks)

[Total: 25 marks]

- Q3. (a) Suppose there are four processes W , X , Y and Z . All clocks run at the same rate (1 unit of time) but initially W 's clock reads 12, X 's clock reads 0, Y 's clock reads 5 and Z 's clock reads 9. At time 12 by W 's clock, W sends a message to X , this message takes 4 units of time to reach X . X then waits one unit of time and then sends a message to Y which takes 2 units of time to reach Y . Later, Y also waits one unit of time and then sends a message to Z . The message reaches Z after 1 unit of time. After Z waits for 2 units of time, a message takes another 1 unit of time to reach back to Y . Assuming that the system implements Lamport's timestamps, draw a picture to illustrate the timestamps for the messages and how the timestamps are obtained. (8 marks)
- (b) Given a scenario of **3 Generals AND 1 Traitor**, find the traitor by using Byzantine Generals algorithm. Show all the steps and diagram necessary to your answer. (9 marks)
- (c) Four clocks W , X , Y and Z are performing internal synchronization by using Berkeley algorithm. W is reading 18:05:00.0 (hr:min:sec) when it is chosen as the master clock. At the same time, X , Y and Z are reading 18:16:00.0, 17:58:00.0 and 18:09:00.0, respectively. Illustrate the clock synchronization process with diagrams from initialization to adjust the time and show the calculations. (8 marks)

[Total : 25 marks]

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**SECTION B**

- Q4. (a) Construct a LAN diagram for a new shopping mall. Your diagram should consider providing internet services to the shopping mall internal staff and customers. Network components in the diagram should include (but not limited to) various types of servers, network switches (hub, router, gateway), and networking equipments (printer, kiosk, computer, mobile devices).

[TIPS: You should group the staff and customer into different subnets]

(11 marks)

- (b) Discuss **THREE (3)** benefits of migrating from a monolithic architecture to an N-tier architecture. (6 marks)

- (c) Provide **THREE (3)** characteristics of cloud architecture that separate it from the traditional one. (6 marks)

- (d) What is the main characteristic that differentiates a software agent from a traditional software program? (2 marks)

[Total : 25 marks]

- Q5. (a) Generate a smart plantation solution (architecture diagram) with Amazon Web Services (AWS) cloud platform. Justify the selection of each cloud service. Your design should consider:

- Deploying sensors for data capturing,
- Storing data for plant growth analysis,
- Generating serverless functions for data processing,
- Notifying farmer for abnormal farm's conditions,
- Having data pipeline for different cloud services,
- Storing sensor log files into the cloud storage,
- Visualizing sensor data for trend analysis, and
- Integrating some relevant intelligent cloud services.

(11 marks)

- (b) Compare the Ricart-Agrawala and Maekawa's Voting algorithms in terms of message complexity, quorum selection, and suitability for distributed mutual exclusion. (6 marks)

UCCD3113 DISTRIBUTED COMPUTER SYSTEMS**Q5. (Continued)**

- (c) Which type of persistence communication can be best applied in transferring a patient's record from his admitted hospital *A* to his physician's hospital *B*? Describe the workflow for utilizing the selected communication service in the scenario above. *[Hints: Both hospitals are located in different cities, and aim to eliminate the traditional overhead associated with operating in-house communication infrastructures.]* (4 marks)
- (d) How does cloud computing enhance the scalability and reliability of streaming services? (4 marks)

[Total : 25 marks]
